A Low-Cost Integrated Motorcycle Theft Detection and Monitoring System Using Arduino, ESP32-CAM, and GSM/GPS Technologies

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***Abstract*—The growing prevalence of motorcycle theft necessitates security systems that extend beyond traditional mechanical locks and alarms. This paper presents a low-cost, modular, and integrated vehicle monitoring and theft detection system utilizing Arduino and ESP32-CAM microcontrollers, GSM/GPS communication, and servo-based visual surveillance. The system incorporates keypad-based ignition authorization, automated camera activation upon unauthorized access, Bluetooth-enabled control, and continuous GPS/GSM location reporting independent of engine state. Experimental validation demonstrates effectiveness in intrusion detection, real-time situational awareness, and reliable location tracking. The solution offers a scalable, affordable, and comprehensive approach to motorcycle security.**

***Keywords*—Motorcycle security, Theft detection system, Arduino–ESP32 integration, GSM/GPS tracking**

I. Introduction

Motorcycle usage has increased rapidly in urban and suburban regions, accompanied by a corresponding rise in vehicle theft. Traditional security mechanisms provide limited protection against modern theft attempts, which often exploit ignition circuit vulnerabilities or bypass audible alarms within seconds. Advancements in embedded electronics—microcontrollers, wireless modules, and low-cost sensors—have enabled smarter vehicle security systems, yet most existing solutions address only isolated components such as GPS tracking, RFID authentication, or camera monitoring. Integrated systems that combine authentication, surveillance, and continuous tracking remain scarce [1]–[4].

This paper presents an integrated vehicle security architecture designed to overcome these limitations. The system employs a keypad-based PIN authentication unit to control ignition access. Incorrect password attempts trigger an ESP32-CAM module mounted on a servo-driven pan mechanism to capture visual evidence of intruders. A Bluetooth interface enables user-controlled camera repositioning and allows remote blocking of the ignition logic in real time. Complementing these features, a covert GSM/GPS subsystem transmits the vehicle's geographic coordinates via SMS every minute, independent of ignition state [1], [3], [6]. The system thereby unifies access control, real-time visual monitoring, and continuous tracking into a single platform that significantly increases theft prevention capability.

II. Methodology

The development of the Vehicle Monitoring and Theft Detection System follows a modular methodology in which the system is divided into four coordinated subsystems: (1) keypad-based access control, (2) video surveillance, (3) servo-based pan mechanism, and (4) GPS–GSM tracking. Each subsystem is implemented on the hardware best suited to its function, and all modules operate together to provide multilayer security.

A. System Overview

The system consists of two cooperating microcontrollers:

1.An Arduino Uno, responsible for ignition authorization, keypad input processing, relay actuation, and Bluetooth command handling.

2.An ESP32-CAM module, serving as the visual monitoring unit while concurrently handling GPS and GSM communication.

By delegating distinct responsibilities to each controller, the system minimizes latency and simplifies firmware complexity, ensuring that surveillance and tracking remain uninterrupted even during user authentication operations [3], [4], [7].

B. Access Control and Authentication Method

1) Keypad Input Processing

A 4×4 membrane keypad forms the primary interface for user authentication. The Arduino continuously scans the keypad in a non-blocking loop, reading input characters and constructing a password string. The system masks entered characters on a 16×2 LCD to maintain user privacy.

2) Verification and State Transition Logic

Upon pressing the confirmation key (#), the Arduino compares the input with the stored 4-digit PIN. The verification result initiates one of four system states:

1.Access Granted: The relay activates to complete the motorcycle ignition circuit, energizing the vehicle [3].

2.Access Denied: The relay remains off, the system logs the failed attempt, and a signal is sent to the ESP32-CAM to capture images or begin streaming [2], [4].

3. System Blocked: In this state, the user can manually block the system at any time using the Bluetooth "B" command. When blocked, the keypad becomes unresponsive and the

ignition circuit is disabled. The system remains in this state until it is unlocked through the Bluetooth "U" command.

4.Reset State: Activatable via Bluetooth, clearing password buffers and restoring the system to its initial state.

C. Intrusion Detection and Camera Activation

When access is denied, the Arduino sends a trigger signal to the ESP32-CAM. The camera responds by serving a live video stream over its built-in HTTP server. The activation logic ensures Immediate visual documentation of the intruder, minimal delay between failed authentication and camera operation and continuous availability of a real-time feed for remote monitoring [2], [4].

D. Servo-Based Pan Mechanism

To enhance visual coverage, the ESP32-CAM is mounted on an SG90 servo motor capable of sweeping through 0°–180°. Two operating modes are implemented: 1.Automatic Sweep Mode: Activated upon incorrect PIN entry. The servo scans horizontally to capture wider surroundings, looping until overridden.

2.Manual Bluetooth Override: Commands ("L", "R", "C") sent through the HC-05 module allow the owner to adjust the camera orientation in real time.

This combination provides maximum situational awareness during intrusion events [2].

E. GPS and GSM Communication Method

A NEO-7M GPS receiver operates continuously to acquire real-time location coordinates, which are parsed using the TinyGPS++ library on the ESP32. A SIM800L GSM module then transmits these coordinates to the owner via SMS at one-minute intervals. This subsystem is designed to function autonomously and remains active regardless of the vehicle's ignition state, thereby providing an additional layer of security even if an intruder bypasses or tampers with the ignition system [1]. The GPS module is interfaced with the ESP32 through UART2. The ESP32 continuously reads and validates NMEA sentences to ensure that only accurate and updated coordinate values are used for reporting. The GSM module sends SMS notifications using standard AT commands. Each message contains the current latitude and longitude. Because the GSM module operates independently of the ignition system, location reporting continues uninterrupted even if the motorcycle is moved without being started, further strengthening the system's theft-detection capability [1], [3].

F. E. ESP32-CAM Integration and Control Logic

The ESP32-CAM is used for live video streaming and intruder capture. On wrong password entry, the Arduino sends a command to the ESP32-CAM.The ESP32-CAM, configured in Access Point mode with a fixed IP address, activates and provides a live video feed accessible through any Wi-Fi–capable device.The servo performs automated 180° sweeps to widen surveillance coverage.Bluetooth commands allow manual adjustment of the servo position.This ensures real-time visual monitoring when suspicious activity is detected. The algorithm of the overall system is presented.

Algorithm 1 Integrated Vehicle Monitoring and Theft Detection System

Require: Keypad input, Bluetooth commands, GPS NMEA data, HTTP client
Ensure: Relay state, servo motion, SMS alerts, camera stream

Initialization
Init keypad, LCD, relay, servo, Bluetooth
Init ESP32-CAM (pins, format, HTTP server)
Init GPS (NEO-7M) and GSM (SIM800L)
$access \leftarrow false$, $blocked \leftarrow false$, $attempts \leftarrow 0$
Display "Enter Password"; relay OFF

Access Control Module (Arduino UNO)
while system powered **do**
  if $blocked = true$ **then**
    Display "SYSTEM BLOCKED"; relay OFF
    Process Bluetooth commands; **continue**
  end if
  Read keypad key
  if key = $*$ **then**
    Clear password buffer
  else if key = # **then**
    if inputPassword = correctPassword **then**
      $access \leftarrow true$; relay ON
      Display "Access Granted"; $attempts \leftarrow 0$
    else
      $access \leftarrow false$; relay OFF
      Display "Access Denied"; send BT alert
      $attempts \leftarrow attempts + 1$; servo sweep
      if $attempts \geq 3$ **then**
        $blocked \leftarrow true$
      end if
    end if
  else
    Append key to buffer
  end if

  Bluetooth Command Handling
  if cmd = "L" **then**
    Servo $\leftarrow$ 30° left
  end if
  if cmd = "R" **then**
    Servo $\leftarrow$ 30° right
  end if
  if cmd = "C" **then**
    Servo $\leftarrow$ 90° center
  end if
  if cmd = "B" **then**
    $blocked \leftarrow true$; relay OFF
  end if
  if cmd = "U" **then**
    $blocked \leftarrow false$; reset LCD
  end if
  if cmd = "S" **then**
    Reset password mode
  end if

  ESP32-CAM Intruder Visualization
  if access = false OR blocked = true **then**
    Activate HTTP camera server
    Serve routes:
    - "/" $\rightarrow$ HTML canvas viewer
    - "/capture" $\rightarrow$ RGB565 buffer
    Continue servo sweep until BT interrupt
  end if

  GPS + GSM Location Tracking
  if GPS fix valid **then**
    Get (lat, lon); format SMS
    Send SMS via SIM800L (AT commands)
  else
    Retry GPS read
  end if
  Wait 60 seconds

  Integration Logic
  if $access = true$ AND $blocked = false$ **then**
    relay ON
  else
    relay OFF
  end if
end while

The operational workflow of the GPS–GSM tracking subsystem is illustrated in Fig. 1. The flowchart outlines the sequence of actions performed by the system, including GPS data acquisition, validation of location accuracy, encoding of coordinates, and periodic SMS transmission to the user. This process ensures continuous monitoring of the vehicle's position.

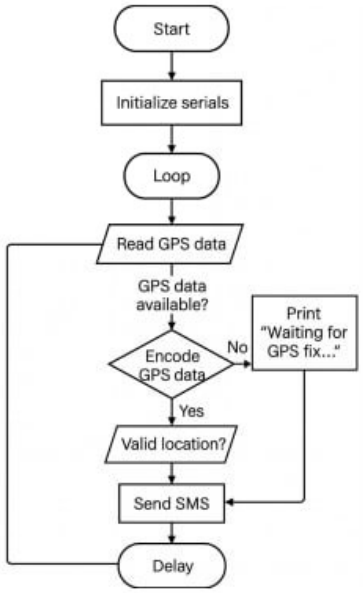


Fig. 1. Flowchart of the GPS data acquisition and SMS reporting process.

The hardware implementation of the proposed system is shown in Fig. 2. The images illustrate how the keypad module, electrical control circuit, ESP32-CAM, GPS receiver, and GSM module are physically integrated into a single prototype. This setup demonstrates the complete functional assembly used for authentication, monitoring, and real-time tracking.

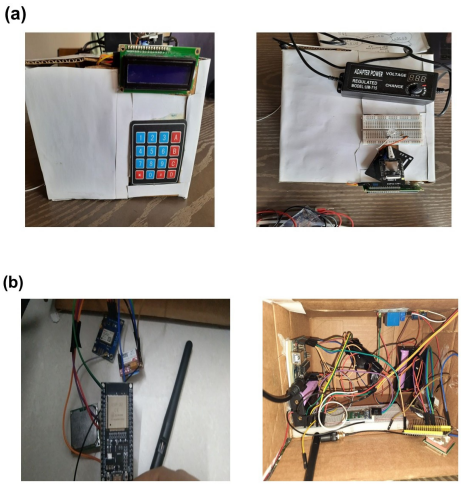


Fig. 2. Overall implementation of the prototype showing hardware integration of the keypad, electrical circuit, ESP32-CAM, GPS, and GSM modules

III. Results and Analysis

The integrated Vehicle Monitoring and Theft Detection System was implemented using a dual-controller architecture, with the Arduino handling authentication and relay control and the ESP32 managing camera streaming, servo motion, and GPS/GSM communication. Experimental testing confirmed that all modules performed consistently and met the intended functional requirements.

A. Authentication and Ignition Control Performance

The 4×4 keypad reliably registered user inputs and transitioned correctly among the four operational states: *Reset*, *Access Granted*, *Access Denied*, and *System Blocked*. Correct password entry enabled the relay and allowed ignition, while incorrect attempts prevented activation and triggered the intrusion-response sequence. These results validate the stability of the keypad-scanning logic and relay-based immobilization [3].

The hardware setup used for the authentication and ignition control subsystem is shown in Fig. 3. The images illustrate the 4×4 keypad interface, password entry module, and relay-based ignition circuit responsible for granting or denying engine access. This configuration demonstrates how user input is processed and how the system enforces secure ignition control.

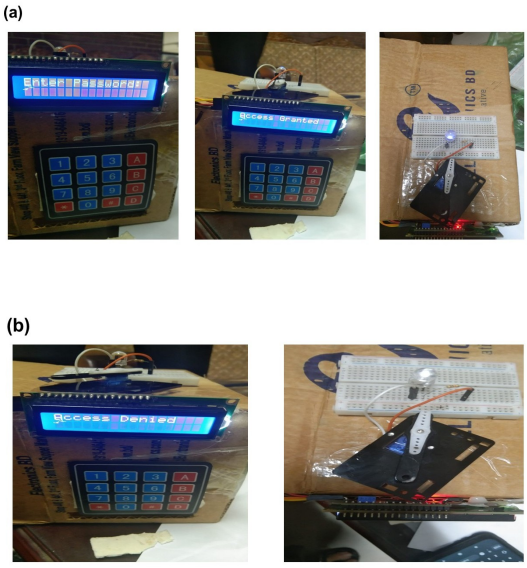


Fig. 3. Implementation of the authentication and ignition control subsystem showing the 4×4 keypad interface, password entry module, and relay-based ignition control hardware in configurations (a) and (b).

B. Intrusion Response and Camera Operation

Wrong password attempts successfully activated the ESP32-CAM subsystem. The servo-mounted camera executed a smooth 0–180° automated sweep, and the live video feed was served without interruption. Bluetooth override commands provided immediate manual adjustment of the camera position, demonstrating responsive integration of mechanical and visual functions [2], [4].

The intrusion-response behavior of the system is illustrated in Fig. 4. When an incorrect password is entered, the ESP32-CAM automatically activates and streams a live video feed, enabling real-time visual monitoring of the intruder. This demonstrates the system's ability to provide immediate surveillance and support rapid user intervention.

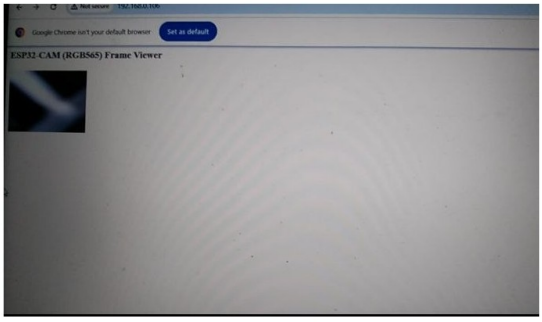


Fig. 4. Live video feed generated by the ESP32-CAM during intrusion detection, demonstrating real-time streaming and automated camera activation.

C. BlueStooth Command Responsiveness

Bluetooth control performed reliably throughout testing. Commands such as "B" and "U" correctly triggered system block and unblock actions, while directional commands adjusted the servo position with minimal latency. This responsiveness is essential for real-time owner intervention [2].

The practical operation of the Bluetooth control interface is shown in Fig. 5. The images demonstrate how the user can issue commands to adjust the servo position or block and unblock the system remotely through a mobile device. This confirms the responsiveness and reliability of the Bluetooth link during real-time interaction with the security system.



Fig. 5. Bluetooth command responsiveness demonstration showing servo control and system block/unblock operations via the mobile interface in views (a) and (b).

D. GPS/GSM Location Tracking Performance

The GPS–GSM subsystem transmitted periodic SMS updates containing the vehicle's coordinates every 60 seconds. Location reporting continued independently of ignition status, confirming that the tracking function remains active even if the motorcycle is moved without being started [1], [3], [6], [7].

The functioning of the GPS–GSM tracking subsystem is illustrated in Fig. 6. The serial monitor output shows the successful retrieval of GPS coordinates and the periodic transmission of location data via SMS. This confirms that the tracking module operates reliably and continues reporting the vehicle's position even when the ignition is off.

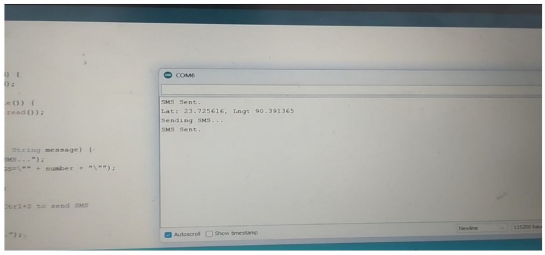


Fig. 6. GPS–GSM tracking output displayed on the serial monitor, showing successful acquisition of GPS coordinates and periodic SMS transmission.

E. Integrated System Behavior

All subsystems operated concurrently without software conflicts. Communication between the Arduino and ESP32 remained stable during state transitions, ensuring correct activation of camera streaming, servo motion, intrusion alerts, and SMS tracking. The overall system behavior confirms that the prototype functions as a coherent, low-cost, and modular anti-theft solution [1]–[3], [5].

F. Comparison With Commercial Anti-Theft Solutions

A comparison with a commercial benchmark, the Spy Spot 4G Hard-Wired GPS Tracker with Kill Switch, highlights the cost advantage of the proposed design. While the commercial device requires a higher upfront cost plus recurring subscription fees, the prototype achieves similar core features—including ignition control, remote blocking, live monitoring, and continuous GPS reporting—at a one-time cost of 5390 BDT. Bulk production can reduce this to approximately 3800 BDT, making the system significantly more affordable for widespread use [1], [4].

IV. CONCLUSION

This paper presented a comprehensive vehicle monitoring and theft detection system that integrates password-based access control, real-time video surveillance, servo-driven camera panning, Bluetooth-based override commands, and continuous GPS–GSM location reporting. By distributing responsibilities between an Arduino Uno and an ESP32-CAM, the system ensures reliable, non-interfering operation across all modules. Experimental implementation confirmed that correct password authentication consistently enables the ignition circuit, while incorrect attempts trigger immediate intrusion response through automated camera activation and panoramic scanning. Simultaneously, the GPS–GSM subsystem transmits updated coordinates every minute, independent of ignition state, ensuring uninterrupted tracking even if the vehicle is physically displaced or tampered with. These coordinated features establish multiple security layers—prevention, detection, monitoring, and recovery [1]–[4].

Overall, the prototype demonstrated effective performance and practical viability using readily available, low-cost components. The modular design makes the system adaptable to

a wide range of vehicle types and enables straightforward upgrades. Future enhancements may include improved camera stabilization, cloud-based video storage, a dedicated mobile application, and integration of modern communication technologies such as 4G/5G. Additionally, artificial intelligence (AI) could further elevate system capability through intelligent motion detection, anomaly analysis, and automated intruder identification, enabling the system to transition from reactive defense to predictive and context-aware security.

Despite these potential extensions, the implemented solution already provides a robust, affordable, and scalable foundation for reliable motorcycle theft prevention, addressing critical security challenges in both urban and suburban environments [1]–[7].

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